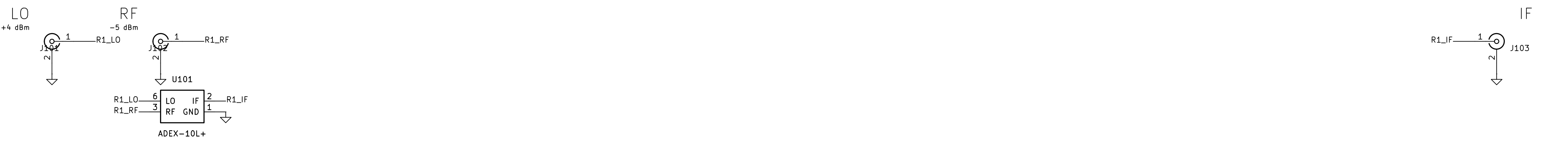
R1 diode mixer, ADEX-10L+

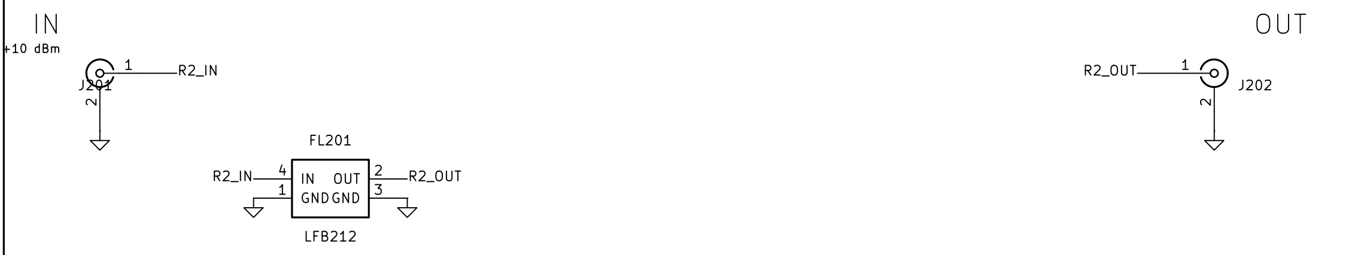
cells (0,0), (1,0)


LO
+4 dBm
RF
−5 dBm
IF
J101
J102
J103
R1_LO
R1_RF
R1_IF
U101
LO
IF
RF
GND
ADEX-10L+

EXPECT: about 7 dB of conversion loss from 10 MHz to 1 GHz with +4 dBm of LO (R10 at its top power), IF from DC to 800 MHz, 60 dB L-R isolation
Mini-Circuits ADEX-10L+: two wideband baluns and a ring of four Schottky diodes (drawn beside it on the silk). The LO (+4 dBm, +1 to +7) turns the diodes on in pairs, so the RF is multiplied by +-1 at the LO rate: the IF is RF x square wave, the sum and difference plus odd LO harmonics. Passive and DC-coupled at IF (a phase detector when RF and LO share a frequency); 1 dB compression at +1 dBm of RF, IP3 +16 dBm. Compare R7/R8 (ADL5350) and AMOD-A's A9 (MC1496).

R2 2.45 GHz filter

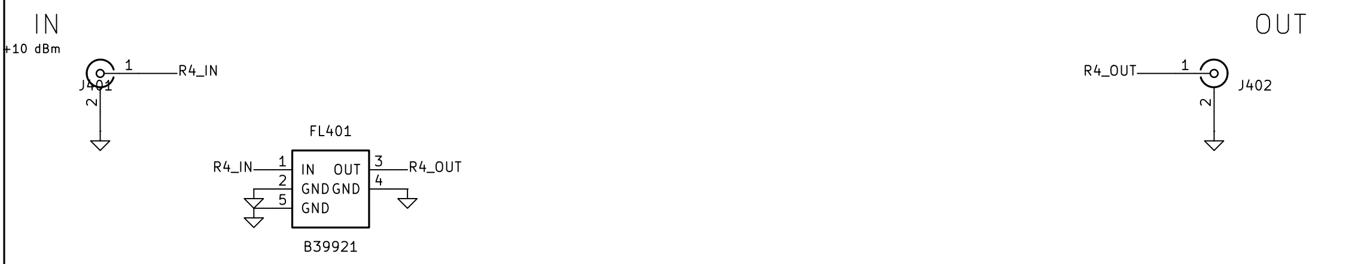
cell (2,0)


IN
+10 dBm
OUT
R2_IN
R2_OUT
J201
J202
FL201
IN
OUT
GND GND
LFB212

EXPECT: 1.7 dB of loss from 2.4 to 2.5 GHz, 30 dB or more below 2.2 GHz and above 2.7 GHz (the datasheet); 50 ohm in and out
Murata LFB212G45SG8A127, a 2012-size LTCC band-pass. The line to each jack is 50 ohm coplanar waveguide (0.29 mm wide, 0.25 to the ground pour) with a via fence; the jack's centre pin has no plane under it (a 2.2 mm void).
Sweep it with R10 or R13 into a power meter, or with a DAC harmonic comb and the ADC.

R4 915 MHz filter

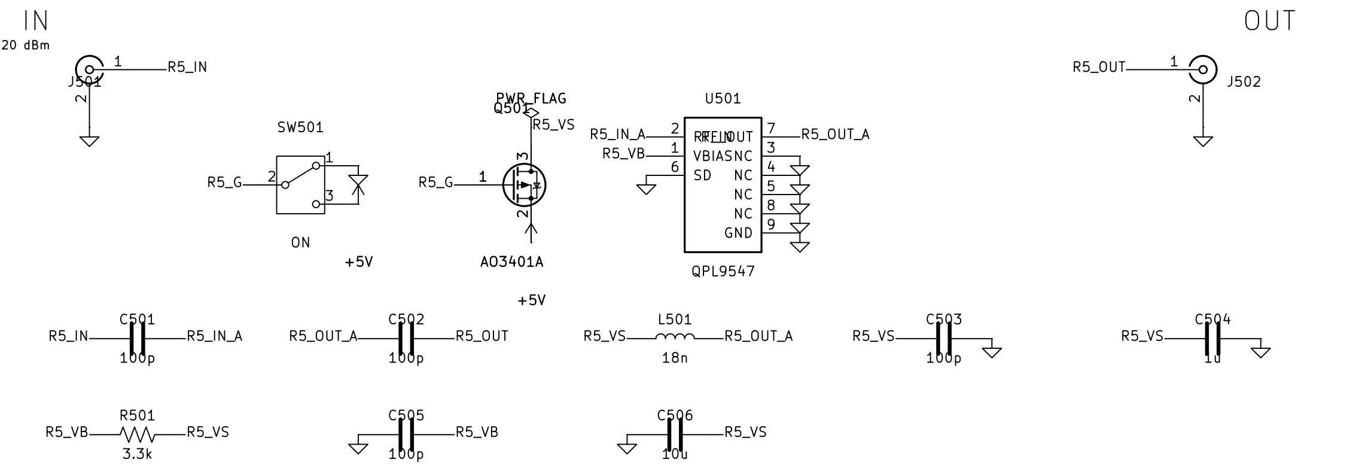
cell (3,0)


IN
+10 dBm
OUT
R4_IN
R4_OUT
J401
J402
FL401
IN
OUT
GND GND
GND
B39921

EXPECT: 1.1 dB of loss from 902 to 928 MHz, 40 dB or more 30 MHz outside the band (the datasheet); 50 ohm in and out
RF360 B39921B2672P810: a surface-acoustic-wave filter for the 915 MHz ISM band, 26 MHz wide, in a 1411 package; its skirts are much steeper than the LTCC's. Lines, fences and launch voids as on R2.

R5 LNA, QPL9547

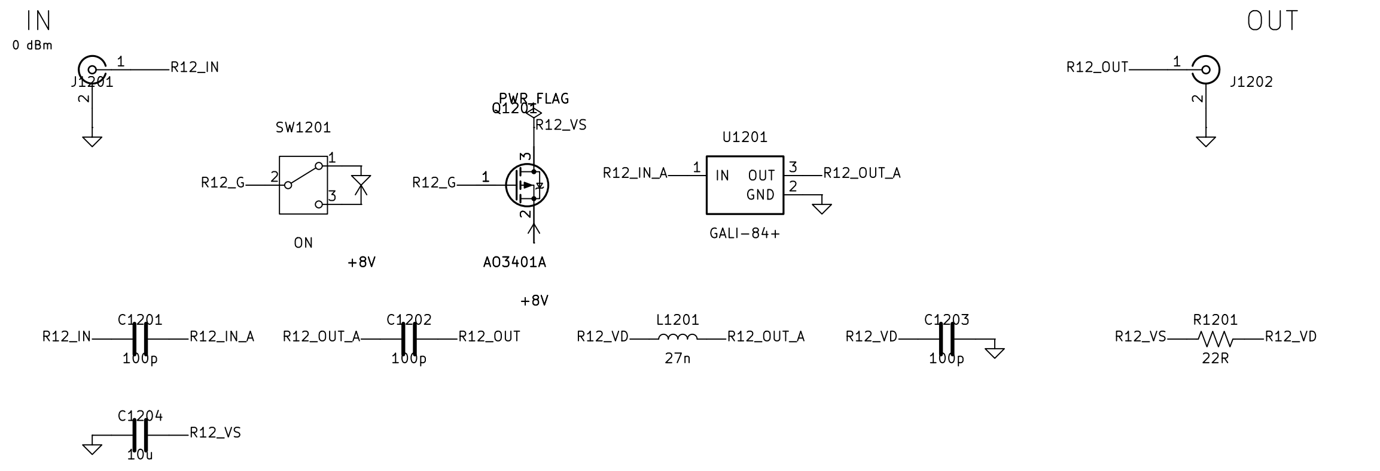
cell (0,1)


IN
−20 dBm
OUT
R5_IN
R5_OUT
J501
J502
SW501
ON
+5V
PWR_FLAG
Q501
R5_G
R5_VS
AO3401A
U501
QPL9547
R5_IN_A
R5_VB
R5_OUT_A
C501
100p
C502
100p
C503
100p
C504
1u
C505
100p
C506
10u
L501
18n
R501
3.3k
R5_VB
R5_VS

EXPECT: 19.5 dB of gain and a 0.3–0.5 dB noise figure from 0.5 to 3 GHz (datasheet: 0.3 dB at 1.9 GHz, OIP3 +39 dBm); 65 mA from 5 V
Qorvo QPL9547: a GaAs pHEMT LNA, internally matched to 50 ohm, 0.1–6 GHz. It sets the noise figure of every receiver here, so put it first (before the filter) unless a strong neighbour overloads it; +39 dBm OIP3 makes that rare.
The 18 nH wire-wound choke (Qorvo's value) feeds 5 V into the output pin; 3.3 k from 5 V to Vbias sets 65 mA.
Never cable a transmitter into it without 30 dB of attenuation (22 dBm absolute maximum at the input).

R12 gain block, GALI-84+

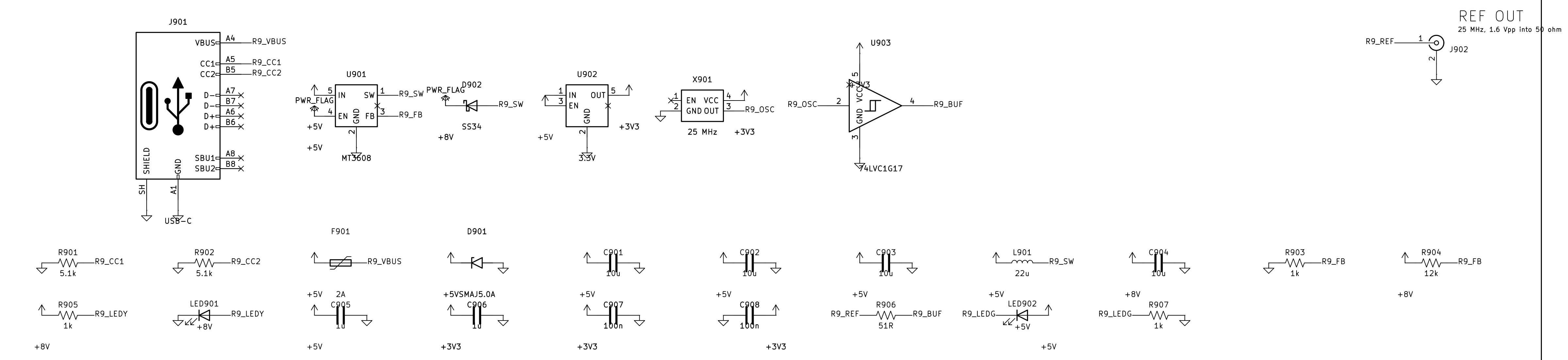
cell (1,1)


IN
0 dBm
OUT
R12_IN
R12_OUT
J1201
J1202
SW1201
ON
+8V
PWR_FLAG
Q1201
R12_G
R12_VS
AO3401A
U1201
IN OUT
GND
GALI-84+
R12_IN_A
R12_OUT_A
C1201
100p
C1202
100p
C1203
100p
C1204
10u
L1201
27n
R1201
22R
R12_VD
R12_VS

EXPECT: 20 dB of gain from 50 MHz to 2.5 GHz (datasheet: 21.9 dB at 1 GHz, 1 dB compression at +20 dBm out); 97 mA from 7.8 V
Mini-Circuits GALI-84+: a Darlington MMIC that sets its own current; the 22 ohm from the 7.8 V rail drops to its 5.8 V device voltage (0.2 W in the 2512). The 27 nH choke (415 ohm at 2.45 GHz, 155 at 915 MHz) feeds the output pin; 100 pF blocks keep the DC off the jacks. Keep the input under 0 dBm for linear gain. The slide switch and P-FET gate the tile's 7.8 V as on AMOD-A.

R9 power, USB-C 5 V in: +5, +7.8 V; 25 MHz reference

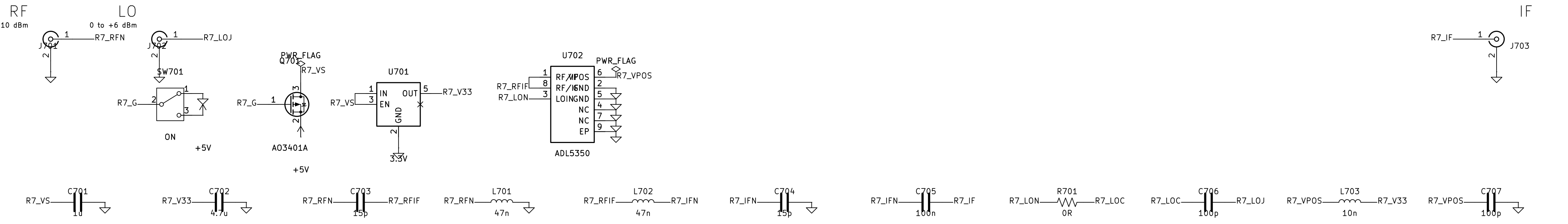
cells (2,1), (3,1)


REF OUT
25 MHz, 1.6 Vpp into 50 ohm
R9_REF
J902
J901
USB-C
R9_VBUS
R9_CC1
R9_CC2
U901
MT3608
R9_SW
R9_FB
PWR_FLAG
D902
SS34
+5V
+8V
U902
+3V3
X901
25 MHz
R9_OSC
U903
74LVC1G17
R9_BUF
R901
5.1k
R9_CC1
R902
5.1k
R9_CC2
F901
R9_VBUS
D901
+5V
2A
+5VSMAJ5.0A
C901
C902
C903
C904
C905
1u
C906
1u
C907
100n
C908
100n
L901
22u
R9_SW
R903
1k
R9_FB
R904
12k
R9_FB
R905
1k
R9_LEDY
LED901
R9_LEDY
+8V
R906
51R
R9_REF
R9_BUF
LED902
R9_LEDG
R907
1k
R9_LEDG
+3V3

EXPECT: +5 V from the USB-C (up to 1.5 A), +7.8 V at 250 mA for the gain blocks (yellow LED), a 25 MHz square wave on REF OUT
USB-C power only: 5.1 k on each CC asks any source for 5 V; a 2 A PTC, a 5 V TVS and 20 uF. The MT3608 boosts to 7.8 V (0.6 V x 13 from 12 k / 1 k) for the GALI-84+ tile, R12 (100 mA); the +8V rail is a strip under row 1 that their headers reach with a via. The TPS7A2033 makes 3.3 V for the TC32H4 TCXO (+-2 ppm) and the 74LVC1G17 that drives REF OUT (51 ohm: 1.6 Vpp into 50 ohm, 3.3 Vpp into a synthesizer's unterminated REF IN; R10's REF IN passes it to R13).
The TCXO's pad 1 (enable or no connection on the drawing) floats, as CostasLink's review settled.

R7 TX mixer, ADL5350

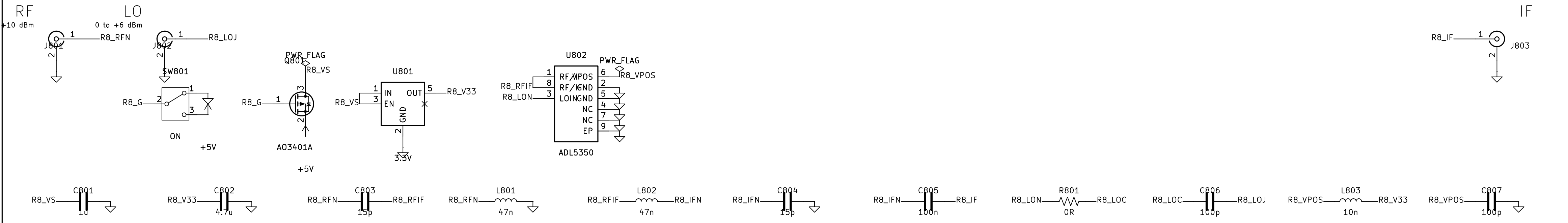
cells (0,2), (1,2)


RF
+10 dBm
LO
0 to +6 dBm
IF
R7_RFN
R7_LOJ
R7_IF
J701
J702
J703
SW701
ON
+5V
PWR_FLAG
Q701
R7_G
R7_VS
AO3401A
U701
R7_V33
3.3V
U702
ADL5350
R7_RFIF
R7_LON
R7_VPOS
C701
1u
C702
4.7u
C703
15p
L701
47n
L702
47n
C704
15p
C705
100n
R701
0R
C706
100p
L703
10n
C707
100p

EXPECT: RF x LO to IF (or IF x LO to RF) with 7–9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1–30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band-tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3–8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up-converts a DAC tone (IF in, RF out) or down-converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with R2 or R4.

R8 RX mixer, ADL5350

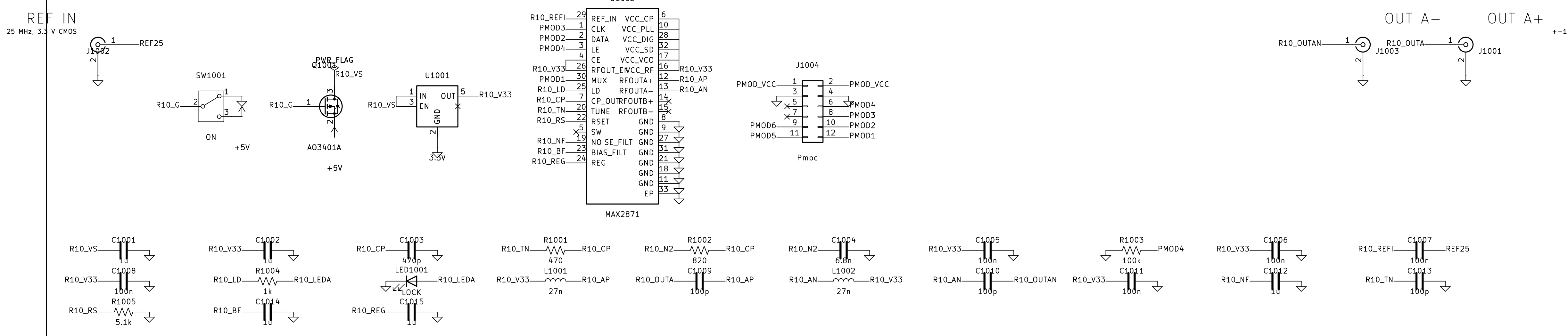
cells (2,2), (3,2)


RF
+10 dBm
LO
0 to +6 dBm
IF
R8_RFN
R8_LOJ
R8_IF
J801
J802
J803
SW801
ON
+5V
PWR_FLAG
Q801
R8_G
R8_VS
AO3401A
U801
R8_V33
3.3V
U802
ADL5350
R8_RFIF
R8_LON
R8_VPOS
C801
1u
C802
4.7u
C803
15p
L801
47n
L802
47n
C804
15p
C805
100n
R801
0R
C806
100p
L803
10n
C807
100p

EXPECT: RF x LO to IF (or IF x LO to RF) with 7–9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1–30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band-tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3–8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up-converts a DAC tone (IF in, RF out) or down-converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with R2 or R4.

R10 synthesizer, MAX2871

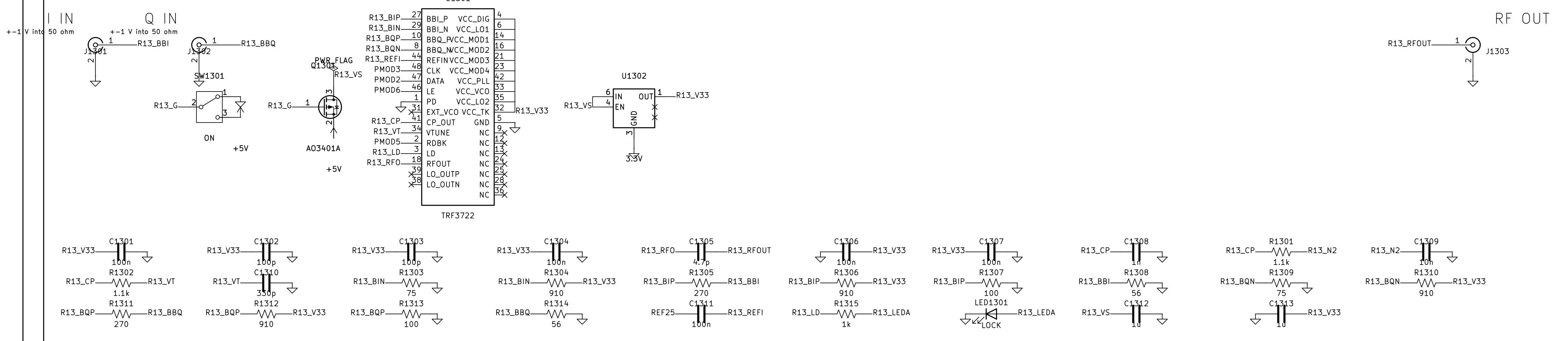
cells (0,3), (1,3)


REF IN
25 MHz, 3.3 V CMOS
REF25
J1001
OUT A−
OUT A+
J1002
J1003
R10_OUTAN
SW1001
ON
+5V
PWR_FLAG
Q1001
R10_G
R10_VS
AO3401A
U1001
R10_V33
3.3V
U1002
MAX2871
REF_IN
CLK
DATA
LE
CE
RFOUT_EN
MUX
LD
CP_OUT
TUNE
RSET
SW
NOISE_FILT
BIAS_FILT
REG
VCC_CP
VCC_PLL
VCC_DIG
VCC_VCO
VCC_RF
RFOUTA+
RFOUTA−
RFOUTB+
RFOUTB−
GND
EP
J1004
Pmod
PMOD_VCC
PMOD1
PMOD2
PMOD3
PMOD4
PMOD5
PMOD6
C1001
1u
C1002
1u
C1003
470p
R1001
470
R1002
820
C1004
C1005
C1006
C1007
100p
C1008
100n
R1004
1k
LED1001
LOCK
L1001
27n
C1009
100p
L1002
27n
C1010
100p
C1011
100n
R1003
100k
C1012
C1013
100p
R1005
5.1k
C1014
1u
C1015
1u

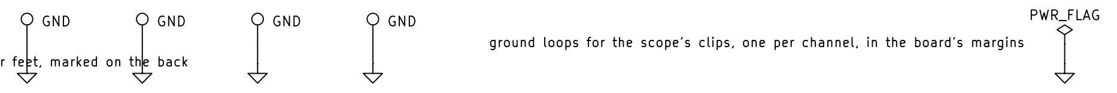
EXPECT: any frequency from 23.5 MHz to 6 GHz on OUT A+ and OUT A− (each −4 to +5 dBm in steps, 180 degrees apart), locked to the 25 MHz on REF IN; the red LED says locked
MAX2871 as CostasLink built it: fractional-N PLL with four VCOs, programmed over SPI from the Pmod (SCLK, SDATA, LE; MUXOUT back). The loop filter is third order, 80–120 kHz with 61 degrees of margin for both 915 MHz (VCO / 4) and 2.4 GHz (VCO / 2) at ICP 2.56 mA (RSET 5.1 k, CP = 7). RFOUTA+ and RFOUTA− each feed a jack through 27 nH to 3.3 V and 100 pF: two outputs, 180 degrees apart; put a 50 ohm terminator on one that is unused. RFOUTB is open and must stay disabled. CE and RFOUT_EN are tied high, LE has 100 k down. REF IN is unterminated; a back trace takes it on to R13.

R13 I/Q transmitter, TRF3722

cells (2,3), (3,3)


I IN
+−1 V into 50 ohm
Q IN
RF OUT
R13_BBI
R13_BBQ
R13_RFOUT
J1301
J1302
J1303
SW1301
ON
+5V
PWR_FLAG
Q1301
R13_G
R13_VS
AO3401A
U1301
TRF3722
BBI_P
BBI_N
BBQ_P
BBQ_N
REFIN
CLK
DATA
LE
PD
EXT_VCO
CP_OUT
VTUNE
RDBK
LD
RFOUT
LO_OUTP
LO_OUTN
VCC_DIG
VCC_LO1
VCC_MOD1
VCC_MOD2
VCC_MOD3
VCC_PLL
VCC_VCO
VCC_LO2
VCC_TK
VCC_MOD4
GND
NC
U1302
R13_V33
3.3V
C1301
100n
C1302
100p
C1303
100p
C1304
100n
C1305
C1306
C1307
C1308
C1309
C1310
330p
C1311
C1312
C1313
R1301
1.1k
R1302
1.1k
R1303
75
R1304
910
R1305
270
R1306
910
R1307
100
R1308
56
R1309
75
R1310
910
R1311
270
R1312
910
R1313
100
R1314
56
R1315
1k
LED1301
LOCK
R13_LEDA

EXPECT: I + jQ from two DACs on 0.4–4.2 GHz, about −3 dBm, image-free (−42 dBc untrimmed)
TRF3722 (PLL, VCO, I/Q modulator) on SPI from the Pmod (LE IO6, readback IO5); reference from R10's REF IN. Each DAC: 56 ohm at the jack, 270 ohm to a node held at 0.25 V (+−0.25 V for +−1 V). DC on I alone is a CW carrier.


GND
GND
GND
GND
PWR_FLAG

no mounting holes: four stick-on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/rf_mod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a quadrature transmitter, a synthesizer, three mixers (one a diode ring), an LNA and a gain block, two
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/
Sheet: /
File: rf_mod.kicad_sch

Title: RF-MOD: 915 MHz, 2.4 GHz and a stretch to 5.8 GHz

Size: A1 Date: 2026-10-25 **Rev: A**
KiCad E.D.A. 10.0.7 Id: 1/1